Project Assessment, Group Project: Student Guide

Title	Trading Strategy Plan and/or Key Inputs
Due Date	End of Week 10, Friday 9:00pm 24-Apr-2026
Weighting	30%
Task	Written Report
Word Limit	2,500 Word maximum excluding charts, tables, diagrams, etc.
Submission	Via Moodle

1. Overview & Goals

This group project assessment provides the opportunity to apply quantitative analysis to the development of and/or evaluation of trading strategies. You will work collaboratively within a team to analyse financial data and implement quantitative research, such as conducting backtests and/or developing models, preferably based on real historical intraday data.

1.1. Learning Outcomes Assessed

This assessment addresses the following Course Learning Outcomes (CLOs):

- Demonstrate an in-depth understanding of the principles, practices, and complexities of financial technology and innovation within global financial systems
- Critically analyse and evaluate real financial problems using technological solutions and data driven decision making frameworks
- Collaborate effectively within diverse teams to deliver well-structured and professional group outputs that reflect critical thinking and strategic planning.

1.2. Goals / Objectives

The objectives of this assessment are to:

- Work collaboratively in a team environment to achieve shared goals
- Gain further experience in use of financial market microstructure data, including the ability to identify and select appropriate datasets for analysis
- Develop domain knowledge in an area of personal interest within trading markets
- Display problem solving skills useful in quantitative research such as initiative, creativity and original thinking.

2. Instructions

- **Preparation**: Review key concepts from lectures and readings.
- **Dataset:** You will need to source data to complete the Group Project. If you have trouble identifying a suitable dataset, contact the Course Convenor for further guidance.

- **Submission:**
  - Report – detailing methodology, analysis and conclusions as a PDF
  - Code – any computational scripts, e.g. Python, R or spreadsheets, e.g. Excel, used for background analysis
  - AI Transparency – if AI tools (e.g. ChatGPT, ML models) are used, students must document their usage and provide justification.

In summary, as a team, you will write a report to present your work in a clear and logical structure. While the exact format may vary depending on the topic selected, most projects will typically include the following sections:

1. Introduction and research objective
2. Data description and background analysis
3. Strategy or model development
4. Testing and validation
5. Evaluation of results
6. Model or strategy refinement
7. Critical reflection with recommendations.

Higher marks will be awarded to projects that use logical data driven arguments and display understanding that extends beyond the standard course materials.

3. Assessment:

Step 1: Team Formation & Role Allocation

Students must form teams and submit their group details via Moodle. If no selection is made, automatic allocation will be made by the Course Convenor. You don't need to notify the Course Convenor about your choice of groups unless you have a non-standard request.
It will also be helpful if you allocate roles to all team members, typical teams consist of:

- Trading – usually involves deep financial market and financial product knowledge, e.g. market microstructure, derivatives versus cash products
- Quantitative Analysis – statistical analysis and mathematical modelling
- Technology – in this context will be coding, but usually also includes physical hardware
- Sales – in this context would focus on writing the reports
- Risk – in this context has a specific focus on trading strategy evaluation and refinement.

Step 2: Project Topic Selection

As per the Course Outline the scope for topic selection is quite broad. Students are expected to select topics that allow for meaningful quantitative modelling and empirical testing using historical intraday market data. Typically the Group Project will involve the following:

- Design and backtest a trading strategy using historical intraday market data, as described in Appendix A
- Model and analyse a key market making input, such as client order flow, inventory management or target profit analysis, as described in Appendices B, C and D
- Explore a topic within market microstructure, such as volatility modelling, price prediction or adverse selection analysis, as outlined in Appendix E. This project must include a clearly defined quantitative model and empirical testing using market data.

Step 3: Data Sourcing, Data Processing & Background Analysis

- Obtain a dataset for your topic, then use a tool of your choice (e.g. Excel, Python, R) to clean the data, if applicable
- Perform any necessary background analysis, e.g. statistics & microstructure metrics
- Split the data into an initial sample for backtesting or model development and a separate out of sample testing or validation.

Step 4: Trading Strategy / Model Development

- For Appendix A, construct a set of well defined rules to implement your trading strategy
- For Appendices B, C & D on market making inputs, students must develop and test a formal quantitative model rather than relying solely on descriptive statistics.

Step 5: Backtesting / Model Testing & Out of Sample Testing / Validation

- Using an initial but limited data sample, estimate, calibrate or implement your trading strategy or model using historical data
- Conduct out of sample testing / validation on a separate unseen historical dataset
- Provide an introductory evaluation of results for the in sample and out of sample data
- For projects involving trading strategies this will typically involve backtesting, while projects involving models will typically involve model testing and validation.

Step 6: Strategy / Model Evaluation

- Interpret and evaluate the results obtained in Step 5
- This should focus on whether the strategy or model performs as expected and whether the results are economically meaningful, stable and robust.

Step 7: Strategy / Model Refinement

- Explore ways to refine your trading strategy and / or model based on Step 6 findings
- This may involve optimisation of parameters, incorporation of additional variables, adjusting assumptions or extending the analysis
- Include an explanation of any changes made to your trading strategy / model analysis
- Repeat the backtest and / or model testing after refinement and compare the results with the original implementation.

Step 8: Critical Reflection

- What are the strengths and limitations of your work ? Please include data to support your arguments and/or tables, charts etc.
- Provide your top three (3) recommendations to improve the implementation of your analysis in a real world setting and explain your reasoning.

Step 9: Peer Evaluation

Using the Moodle Team Evaluation Tool, provide feedback on your peers with respect to:

- Engagement and participation, technical contribution, communication and collaboration, Timeliness and reliability, Initiative and critical thinking
- Comment on the contribution, any challenges & possible differentiation for your peers.

Appendix A: Design & Backtest a Trading Strategy

A.1: Team Formation & Role Allocation

This step corresponds to Step 1: Team Formation & Role Allocation in the main assessment instructions.

A.2: Trading Strategy Selection

- Provide a clear statement of your trading strategy and include all key assumptions necessary for implementation
- Please also include an explanation on why you selected the specific trading strategy
- Design and backtest a trading strategy using historical intraday market data

The following questions may provide a systematic way to select your trading strategy:

- What is your 'Edge' ? What areas of markets interest you ?
- What Time Horizon do you prefer ? High, Medium, Low Frequency
- What Asset Class do you find interesting ? What Asset Class do you understand best ? Select between Equities, ETFs, Cryptocurrencies, Fixed Income, Currencies, Commodities, New Economy Markets, e.g. Carbon Credits, ESG Strategies
- What Data can you access ? All students have access to Factset
- Which Trading Strategies do you understand best or wish to learn more about ? Market Making, Directional, Arbitrage

A.3: Data Sourcing, Data Processing & Background Analysis

Examples of background analysis may include:

- price behaviour of the instruments, such as trends, volatility and intraday price movements
- bid–ask spreads and market liquidity
- trading volume, trade frequency and turnover
- order book depth and liquidity across different price levels
- short term price dynamics or microstructure patterns that may influence trade entry or exit signals
- volatility measures that may influence position sizing or risk management rules
- any indicators or signals that may be used to support the trading strategy.

Students should consider how the results of the background analysis may influence the design of the trading plan in the next section.

A.4: Trading Plan Development

- Formulate a trading plan for your strategy, the following steps are recommended:
    - Trade Entry Criteria
    - Trade Exit Criteria
    - Position Size Determination
    - Trade Execution Rules.

A.5: Backtesting / Model Testing & Out of Sample Testing / Validation

Examples of backtesting and out of sample testing may include:

- Performance analytics, e.g. total & annualised returns, CAGR, number of trades, win rate, alpha, beta & correlation
- Risk measures, e.g. volatility, average true range (ATR), value-at-risk, expected shortfall, maximum drawdown
- Risk return metrics, e.g. Sharpe, Sortino, Treynor & Information Ratios
- Equity curve of cumulative returns over the test period
- Other metrics, e.g. turnover, capacity, P&L + holding period distributions

Advanced students can attempt forward testing by application of your trading strategy using real time data, without actual execution, to simulate how strategy performance in a live environment and observe operational factors like data lags, execution frequency, and dynamic market conditions.

A.6: Strategy Evaluation

Review the initial analysis from your backtest, then add further

- Performance analytics, as appropriate
- Risk assessment measures
- Risk / Return metrics
- Other metrics, if relevant.

A.7: Strategy Refinement

Explore ways to refine your trading strategy, such as:

- Optimisation of key parameters in the trading plan
- Further research and analysis, i.e. algorithm modifications for
    - More realistic modelling, e.g. transaction costs, short selling restrictions
    - Market announcements, e.g. company earnings, event flags, sentiment analysis
    - Seasonality and/or time of day effects.
    - Price prediction model improvements, e.g. AI / ML / advanced statistics
    - Risk management improvements, e.g. advanced inventory management, adverse selection models, volatility indicators, advanced liquidity measures
- Repeat the backtest with the refined trading strategy, i.e. conduct a new backtest that incorporates any changes to the original trading plan
- Compare the results between your initial and subsequent backtest.

Remember to complete Step 8: Critical Reflection and Step 9: Peer Evaluation.

Appendix B: Client Order Flow Analysis

Understanding the behaviour of client order flow helps market makers estimate:
- expected trading activity and quote execution frequency
- potential inventory accumulation and inventory risk
- the balance between buying and selling pressure in the market.

Remember that market makers care about how frequently they get hit, how large those trades are, and whether the flow is balanced.

B.1: Team Formation & Role Allocation

This step corresponds to Step 1: Team Formation & Role Allocation in the main assessment instructions.

B.2: Research Objective – Client Order Flow

Provide a clear statement of the objective of your analysis and explain why client order flow is important to market making strategies.

The following questions may provide a systematic way to define your research objective:
- How frequently do trades arrive in the market ?
- What is the typical size of trades executed ?
- Is order flow balanced between buying and selling ?
- Does client order flow vary during the trading day ?

Examples of research objectives include:
- estimating the arrival rate of trades
- modelling the distribution of trade sizes
- analysing order flow imbalance between buyers and sellers.

B.3: Data Sourcing, Data Processing & Background Analysis

Examples of background analysis may include:
- number of trades per minute or per hour
- average and median trade size
- distribution of trade sizes
- buy volume versus sell volume
- order flow imbalance over time
- time between successive trades.

B.4: Client Order Flow Model

- Develop a model for client order flow and explain the logic and assumptions of your approach
- At minimum, the project must include a formal model of trade arrival rates, trade size, and order flow imbalance.

Possible modelling approaches include:
- trade arrival rate models
- inter-arrival time models

- trade size distribution models
- directional order flow models.

Students should provide a clear explanation of how the model is constructed and how it relates to the behaviour of client order flow.

B.5: Model Testing & Out of Sample Validation

Examples of model testing and out-of-sample validation may include:

- comparing predicted trade arrival rates against realised trade activity
- comparing predicted trade size distributions against realised trade sizes
- comparing predicted buy / sell imbalance against realised order flow
- testing whether the model performs consistently across different trading days or market conditions.

Advanced students can attempt additional model validation techniques, such as testing whether their model remains stable across different trading days, volatility regimes or market conditions, in order to assess the robustness of the model's ability to describe client order flow behaviour.

B.6: Model Evaluation

Review the initial analysis from your client order flow model and add further evaluation where appropriate. This may include assessment of:

- model accuracy and stability
- robustness across different time periods
- limitations of the modelling approach.

B.7: Model Refinement

Explore ways to refine your client order flow analysis, such as:

- adjusting model parameters
- incorporating additional market variables
- analysing different time periods of the trading day
- refining the classification of buy and sell initiated trades
- testing alternative modelling approaches.

Repeat the model estimation and testing process with the refined approach and compare the results with the original model.

Remember to complete Step 8: Critical Reflection and Step 9: Peer Evaluation.

Appendix C: Inventory Management

Effective inventory management is critical for market making strategies. Market Makers must manage inventory risk while continuing to provide liquidity to the market. Inventory risk often arises from client order flow and changes in market conditions.

Understanding inventory dynamics helps Market Makers estimate:

- how quickly inventory positions may accumulate
- the risk associated with holding positions during volatile market conditions
- the effectiveness of hedging or risk management strategies.

Poor inventory management can lead to significant losses during periods of directional client flow or market stress.

C.1: Team Formation & Role Allocation

This step corresponds to Step 1: Team Formation & Role Allocation in the main assessment instructions.

C.2: Research Objective – Inventory Management

Provide a clear statement of the objective of your analysis and explain why inventory management is important to market making strategies.

The following questions may provide a systematic way to define your research objective:

- How does inventory accumulate through time as trades are executed ?
- How does market volatility affect inventory risk ?
- How can inventory risk be reduced through hedging or quoting behaviour ?
- How does inventory exposure change during periods of market stress ?

Examples of research objectives include:

- modelling the dynamics of inventory accumulation
- estimating the risk associated with holding inventory positions
- analysing the impact of volatility or directional flow on inventory levels.

C.3: Data Sourcing, Data Processing & Background Analysis

Examples of background analysis may include:

- trading volume and trade frequency
- volatility measures such as realised volatility or ATR
- bid–ask spreads and market liquidity
- patterns of directional order flow.

C.4: Inventory Model

Develop a model that describes how inventory evolves through time as trades are executed. At minimum, the project must include a formal model of inventory dynamics that captures how inventory positions change in response to executed trades.

Possible modelling approaches include:

- simple inventory accumulation models based on executed trades
- inventory models incorporating volatility or price risk

- inventory models incorporating directional order flow
- inventory models incorporating hedging through related instruments.

Students should provide a clear explanation of how the inventory model is constructed and how it relates to market making behaviour.

C.5: Model Testing & Out of Sample Validation

Examples of model testing and out-of-sample validation may include:

- comparing predicted inventory paths against realised inventory outcomes
- evaluating inventory risk under different levels of trading activity or volatility
- testing how inventory behaves under periods of directional order flow
- assessing whether hedging or risk management rules reduce inventory exposure.

Advanced students can attempt additional testing scenarios to evaluate the behaviour of the inventory model under simulated stress conditions e.g. sustained directional order flow or elevated volatility to assess inventory risk under extreme market conditions.

C.6: Model Evaluation

Review the initial analysis from your inventory model and add further evaluation where appropriate. This may include assessment of:

- inventory risk under different market conditions
- sensitivity of inventory positions to volatility or directional flow
- limitations of the modelling approach.

C.7: Model Refinement

Explore ways to refine your inventory model, such as:

- incorporating intraday volatility measures, e.g. widen spreads during periods of increased volatility or market stress
- analysing time-of-day effects in inventory accumulation
- incorporating adverse selection or price impact
- incorporating hedging using correlated assets or instruments, e.g. futures
- adjust / skew prices and / or spreads in response to current inventory levels & volatility.

Repeat the model estimation and testing process with the refined approach and compare the results with the original model.

Remember to complete Step 8: Critical Reflection and Step 9: Peer Evaluation.

Appendix D: Target Profit Analysis

Target Profit Analysis is important for market making strategies. Market Makers must decide how to allocate capital across different trading opportunities in order to balance expected profitability and risk.

Understanding the relationship between spreads, trading activity and risk helps Market Makers estimate:

- how much spread revenue may be available across different securities
- how frequently trading opportunities may occur
- the level of risk associated with deploying capital in different instruments.

Target Profit Analysis helps Market Makers identify where capital may be used most effectively across a market, such as Stocks or ETFs. Remember that Market Makers must decide where to deploy capital by balancing spreads, trading activity and risk across different securities.

D.1: Team Formation & Role Allocation

This step corresponds to Step 1: Team Formation & Role Allocation in the main assessment instructions.

D.2: Research Objective – Target Profit Analysis

Provide a clear statement of the objective of your analysis and explain why target profit analysis is important to market making strategies.

The following questions may provide a systematic way to define your research objective:

- How wide are bid ask spreads across different securities ?
- Which securities trade most frequently ?
- How does volatility affect the potential profitability of market making ?
- How should Market Makers allocate capital across securities to maximise expected profit ?

Examples of research objectives include:

- estimating expected spread capture across a range of securities
- analysing the relationship between trading activity and potential execution opportunities
- comparing volatility and risk across a set of instruments
- identifying how capital may be allocated across securities to improve expected profitability.

D.3: Data Sourcing, Data Processing & Background Analysis

Examples of background analysis may include:

- average bid ask spreads
- trading volume and trade frequency
- realised volatility or other risk measures
- turnover and trading activity across securities.

D.4: Target Profit Model

Develop a model to estimate expected profitability across a set of securities. At minimum, the project must include a formal model that combines spread, trading activity and risk to estimate expected profitability and / or capital usage.

Possible modelling approaches include:

- estimating spread capture across different securities
- modelling execution opportunities based on trading activity
- incorporating volatility or risk measures into profitability estimates
- comparing expected profitability across multiple instruments.

Students should provide a clear explanation of how the target profit model is constructed and how it relates to capital allocation decisions.

D.5: Model Testing & Out of Sample Validation

Examples of model testing and out-of-sample validation may include:

- comparing estimated spread capture across different securities
- evaluating execution opportunities based on trading activity
- analysing the relationship between spreads, trading volume and volatility
- testing whether expected profitability remains stable across different time periods.

Advanced students can attempt additional validation by testing if profitability estimates remain consistent across different time periods, different securities or market environments.

D.6: Model Evaluation

Review the initial analysis from your target profit model and add further evaluation where appropriate. This may include assessment of:

- differences in expected profitability across securities
- sensitivity of expected profits to changes in spreads, trading activity or volatility
- limitations of the modelling approach.

D.7: Model Refinement

Explore ways to refine your target profit model, such as:

- incorporating additional securities or asset classes
- analysing different time periods of the trading day
- incorporating inventory risk, hedging costs or transaction costs
- incorporating market impact or capital constraints
- testing alternative modelling approaches
- allocating a fixed amount of capital across a set of securities in order to maximise expected profitability.

Repeat the model estimation and testing process with the refined approach and compare the results with the original model.

Remember to complete Step 8: Critical Reflection and Step 9: Peer Evaluation.

Appendix E: Market Microstructure Topics

Other possible project topics for a Group Project could be focussed on specific areas within market microstructure:

- Intraday Volatility Calculations, e.g. data selection, optimal historical holding period, fundamental vs. transitory volatility, other metrics such as average true range (ATR)
- Intraday Price Prediction Models, e.g. combine Order Book and Trade Feed data or use of Artificial Intelligence (AI) / Machine Learning (ML) models
- Adverse Selection models such as Probability of Informed Trading, e.g. analysis of Trade Feed data using Broker IDs
- Optimal Execution, e.g. when to use market vs. limit orders based on Order Book & Trade Feed data.

If you are interested in exploring any of the above topics, contact the Course Convenor for further information.

Assessment Criteria & Feedback Guidance

Criteria	HD (85-100%)	D (75-84%)	C (65-74%)	P (50-64%)	F (<50%)
Topic Selection, Data Processing & Strategy / Model Implementation (25%) *(Applying & Creating)*	Topic selection is very challenging with clear, logical, well-structured implementation. Uses advanced models and work is explained well.	Topic selection is moderate to advanced. Demonstrates good implementation with some areas for refinement.	Topic selection difficulty is moderate with appropriate implementation. Further explanation or optimisation is required.	Topic selection is basic and implementation lacks clarity in methodology or assumptions. Limited explanation of work.	Topic selection and/or implementation is unclear, lacks coherence, or has methodological flaws. Fails to apply basic models correctly.
Analysis & Evaluation (25%) *(Analysing & Evaluating)*	Executes comprehensive analysis & testing of the trading strategy or model. Uses additional metrics effectively and provides insightful evaluation.	Conducts solid analysis & evaluation, although some improvements could be made.	Performs analysis adequately but either lacks depth and/or has limited evaluation of results.	Provides very basic analysis or analysis contains some errors. Critical evaluation of results is also basic.	No analysis or major flaws in execution. Lacks critical evaluation of results.
Critical Reflection, Refinement & Recommendations (25%) *(Evaluating & Synthesising)*	Provides deep, evidence-based reflections on strengths and limitations. Proposes innovative and feasible improvements.	Offers well-supported critical reflections and recommendations. Some areas for deeper analysis or stronger justification.	Provides reasonable reflection but lacks depth or fails to fully justify recommendations.	Attempts reflection but lacks strong evidence or depth. Recommendations are vague or generic.	Minimal or no reflection. Recommendations are missing, unfeasible, or lack justification.
Group Collaboration & Contribution (25%) *(Applying & Evaluating)*	Demonstrates exceptional teamwork. All members contribute equitably, leveraging diverse skills. Evidence of collaboration, role distribution & support.	Strong teamwork with clear contributions from all members. Some minor imbalances in workload, but overall well-coordinated effort.	Reasonable teamwork with most members contributing. Some imbalances in effort or unclear role distribution.	Limited teamwork. Unequal contribution or poor coordination. Issues with participation affect quality of work.	Lack of teamwork. Significant contribution imbalance, poor communication, or conflicts that impact project success.